

| Course                                                   | Mandatory Activity                                                           | Skills Developed                                                                                      |
|----------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>AI on Cloud Computing</b>                             | AWS ML & MLOps Foundations; Internal Hackathon                               | Cloud AI, AWS, ML deployment, MLOps fundamentals, cloud-based AI development                          |
| <b>Fundamentals of Parallel Processing</b>               | CUDA Toolkit; SIH participation                                              | GPU computing, CUDA, parallel programming, performance optimization, problem solving                  |
| <b>Web Application Development</b>                       | Full-stack with AI Bridge capstone; External hackathon                       | Full-stack development, AI integration, application architecture, deployment, teamwork                |
| <b>Big Data Analytics</b>                                | Coursera: Kafka & Spark for Data Engineers; Mini Project                     | Big data processing, Kafka, Spark, data engineering, real-time analytics, distributed data processing |
| <b>Time Series Forecasting &amp; Predictive Analysis</b> | Microsoft Power BI Data Analyst Professional Certificate; Internal Hackathon | Data analytics, Power BI, visualization, predictive analytics, forecasting, business intelligence     |
| <b>Data Warehousing &amp; Data Mining</b>                | Real-time data warehouse case study; External Hackathon                      | Data warehouse design, real-time data architecture, data mining, analytical thinking, solution design |